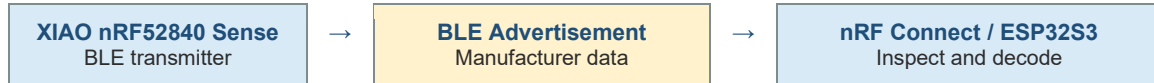




Sending TinyML Results over BLE

1. Communication Goal

After each inference, the nRF52840 Sense broadcasts a small BLE manufacturer-data packet. Students first inspect the packet using nRF Connect, then receive and decode it using the ESP32S3 gateway.



2. BLE Packet Sent by the nRF52840 Sense

The transmitter advertises with the device name Gesture01. The application payload contains seven bytes; nRF Connect and the ESP32S3 also show the two-byte company ID at the beginning.

Displayed byte(s)	Meaning	Example	Notes
0–1	Company ID	FF FF	0xFFFF
2	Protocol version	01	Version 1
3	Node ID	01	Transmitter 1
4	Gesture code	50 / 46 / 57	ASCII P / F / W
5	Selected confidence	5D	93% = 0x5D
6	Punch confidence	5D	0–100
7	Flex confidence	07	0–100
8	Sequence number	04	New result counter

3. Part A — Start the BLE Transmitter

1. Open Classifier_BLE_1.ino. Keep yesterday's model.h in the same folder.
2. Select Seeed Studio XIAO nRF52840 Sense, compile, and upload.
3. Open Serial Monitor at 115200 baud and wait for "BLE data is now being broadcast."
4. **Make sure to change Guesture_1 to your Project name**
5. The initial packet uses gesture code W (waiting). Perform a punch or flex to update the advertisement.

The communication call made after inference is:

```
Advertise_Gesture(gestureCode,  
selectedConfidence,  
punchConfidence,  
flexConfidence);
```

4. Part B — Detect and Decode the Packet with nRF Connect

1. Install/open nRF Connect on a phone with Bluetooth enabled.
2. Select SCANNER and start a scan.
3. Find the device named **under your project name** and open its advertising details.
4. Locate Manufacturer Specific Data. Record the hexadecimal bytes.
5. Perform one punch and one flex. Observe that the gesture byte, confidence bytes, and sequence number change.

Manual decoding: 50 = P (Punch), 46 = F (Flex), and 57 = W (Waiting). Confidence bytes are hexadecimal numbers; for example, 5D hex = 93 decimal.

5. Part C — Receive the Same Packet on the ESP32S3 Gateway

1. Open Gusture_Rec.ino.
2. Select Seeed Studio XIAO ESP32S3. Set USB CDC On Boot to Enabled.
3. Compile and upload, then open Serial Monitor at 115200 baud.
4. The gateway scans for the exact name Gesture01, validates the packet, removes repeated advertisements using the sequence number, and prints each new result.
5. Perform punch and flex gestures and compare the gateway values with the nRF Connect values.

Expected gateway output

```
Module: Gesture01
RSSI: -47 dBm
Gesture code: P
Classification: Punch
Selected confidence: 93%
Punch confidence: 93%
Flex confidence: 7%
Packet number: 4
```